

LABORATORY OF AUTOMATION SYSTEMS

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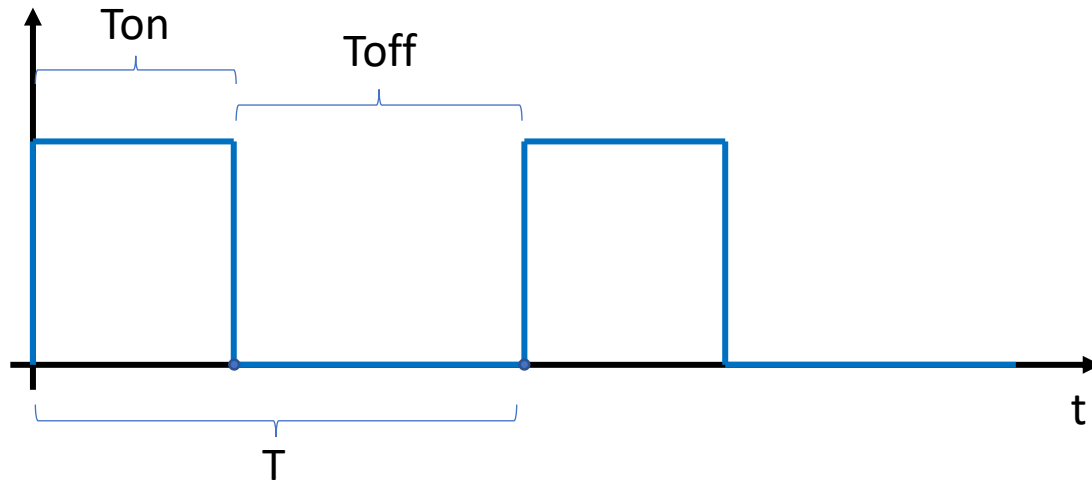
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Pulse Width Modulation (PWM)

Pulse Width Modulation, commonly known as PWM, is a way to regulate the average power delivered to the motor by switching a power signal ON and OFF similar to a square wave.

In a PWM signal we have two parameters: **Frequency** and **duty-cycle**

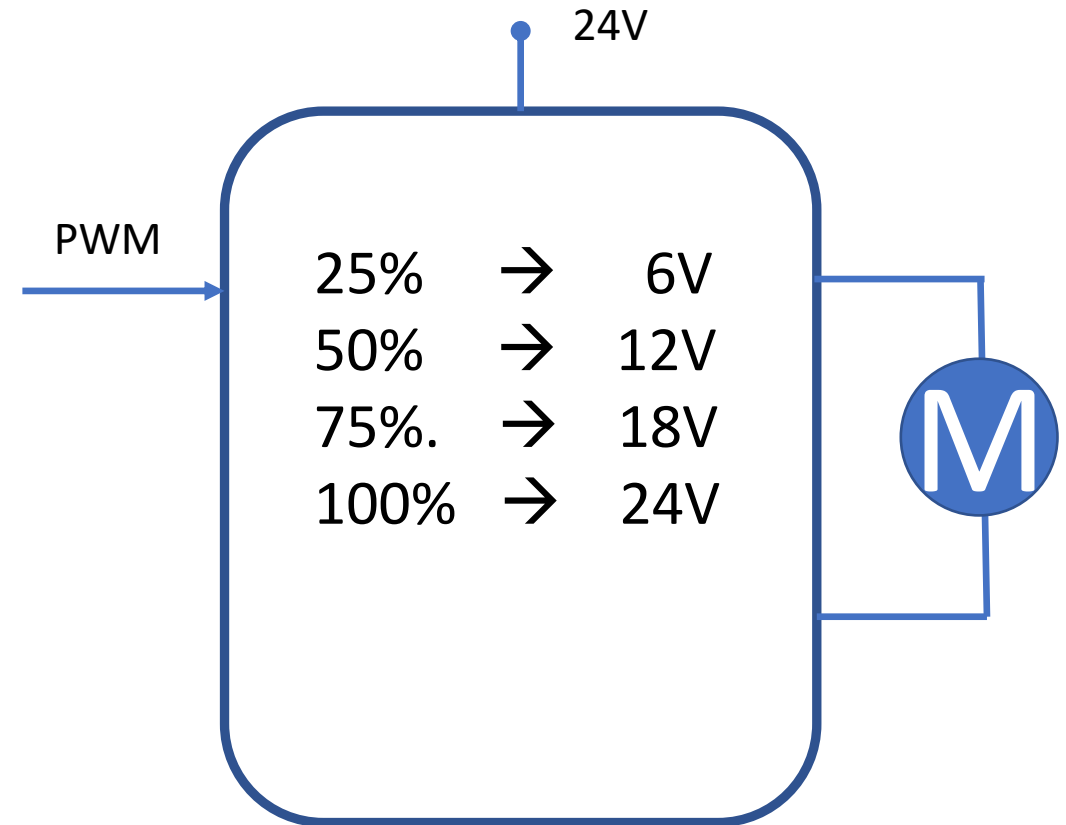
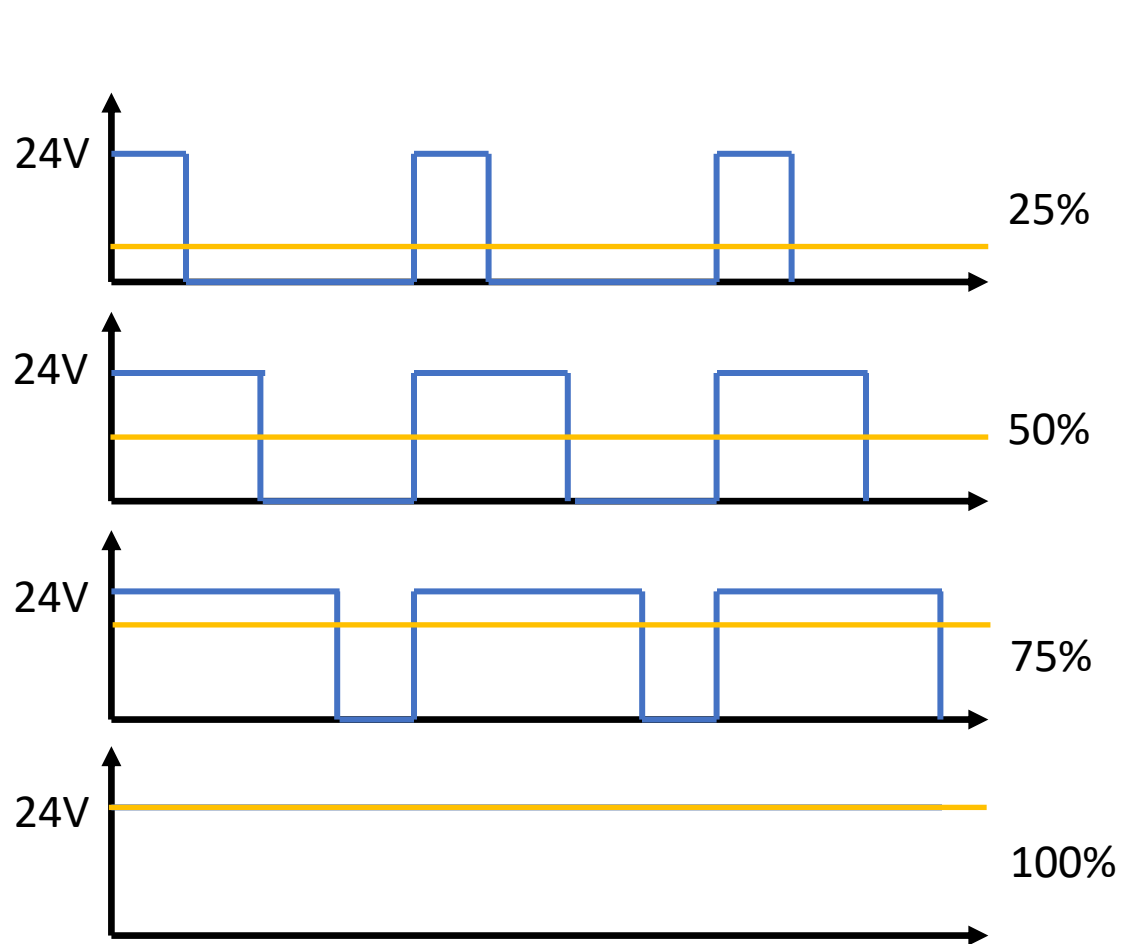


$$period = T_{on} + T_{off} = T$$

$$frequency = \frac{1}{T}$$

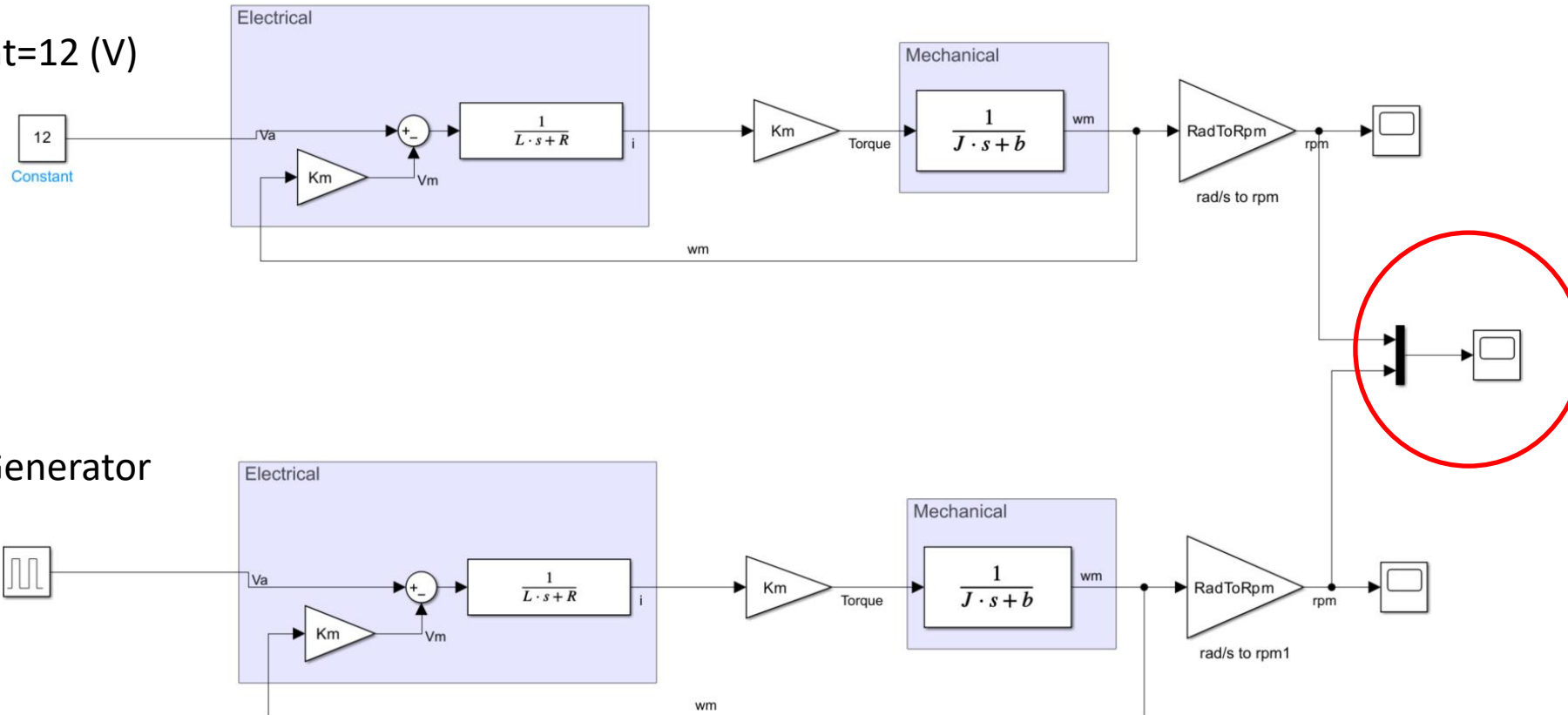
$$duty cycle = \frac{T_{on}}{T_{on} + T_{off}} * 100 = \frac{T_{on}}{T} * 100$$

Pulse Width Modulation (PWM)

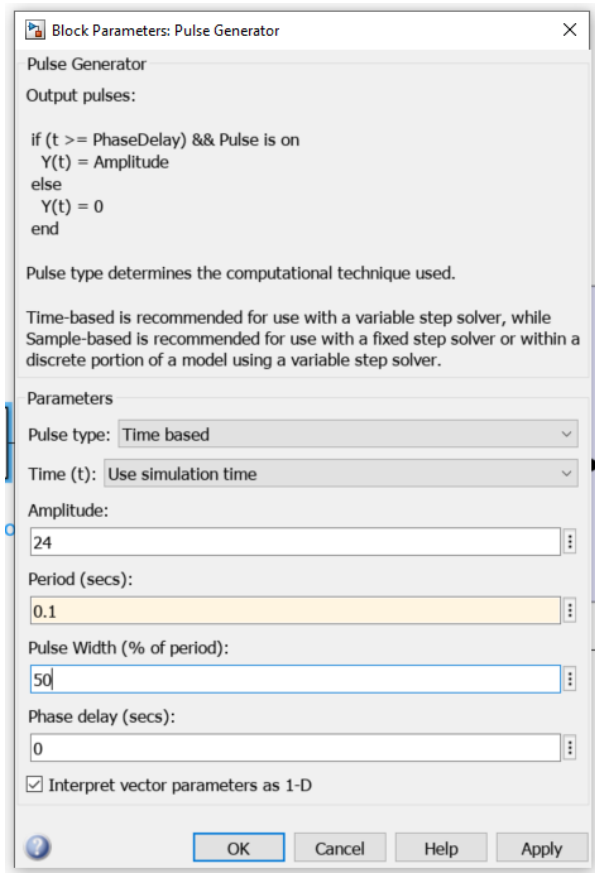


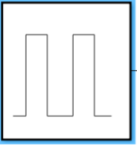
Pulse Width Modulation (PWM)

Constant=12 (V)



Pulse Width Modulation (PWM)



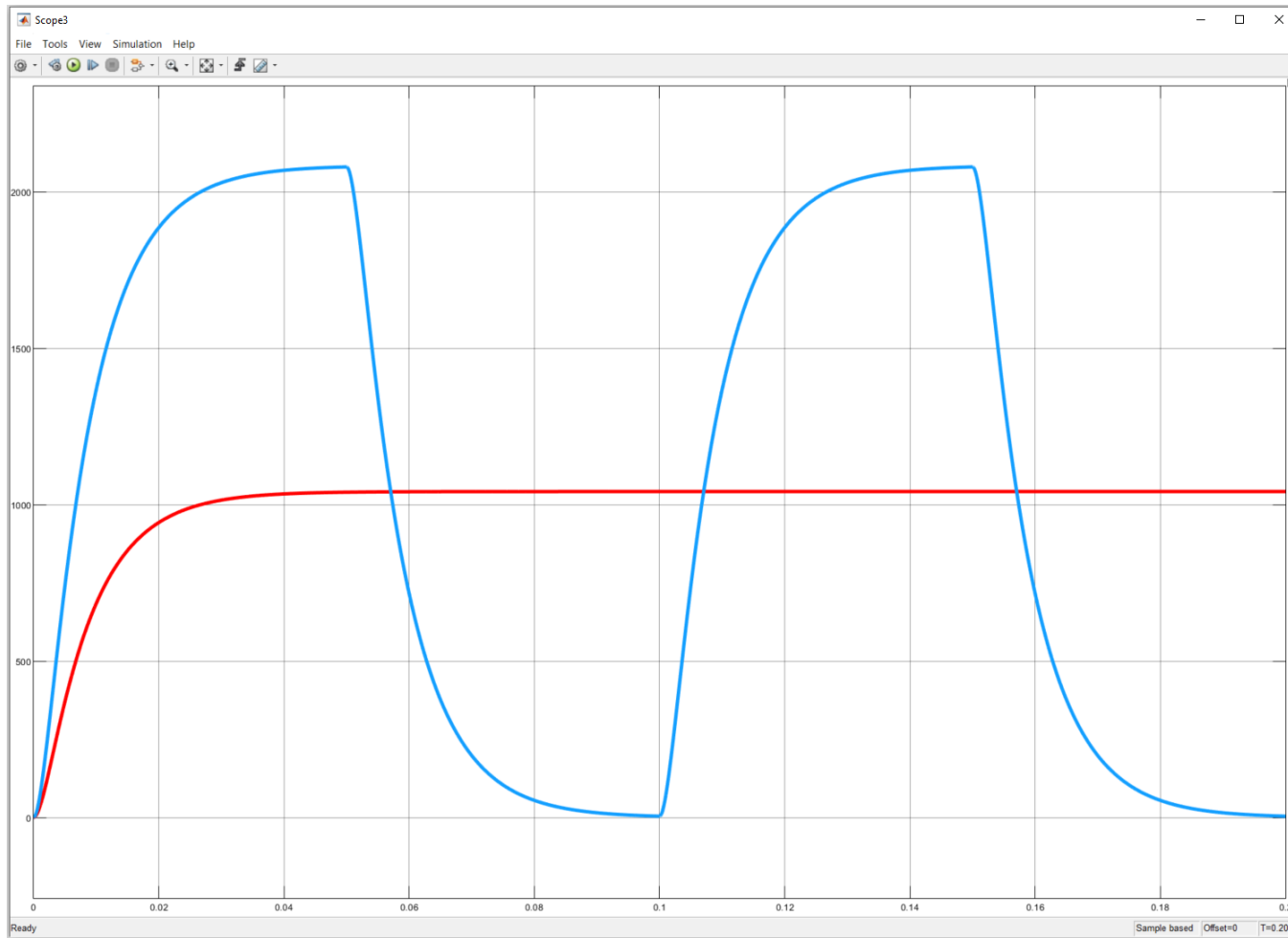

Pulse
Generator

Try with different **Period (secs)**:

- 0.1
- 0.01
- 0.001

Simulink Stop Simulation: 0.2

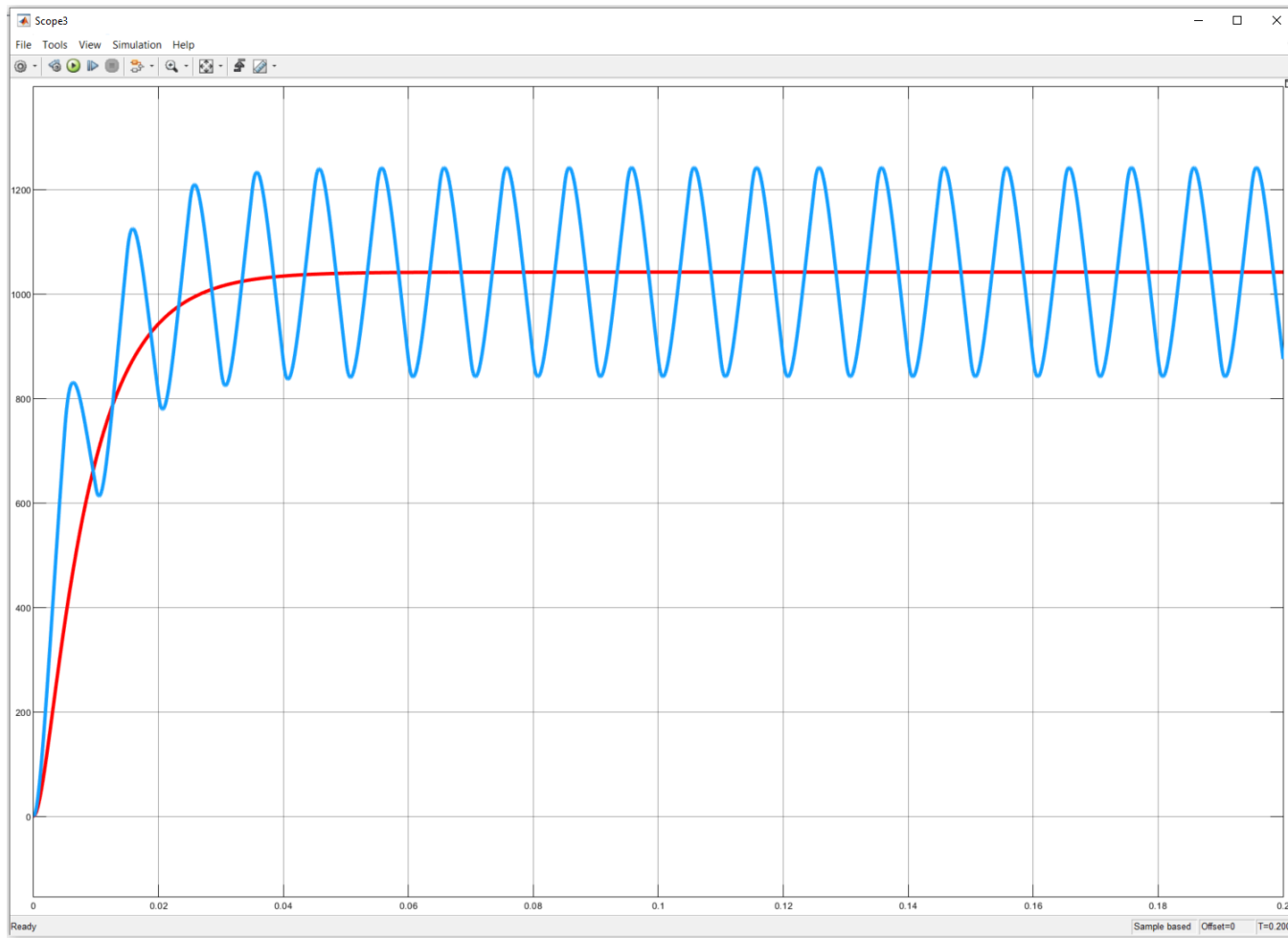
Pulse Width Modulation (PWM)



Period (secs): 0.1

Simulink Stop Simulation: 0.2

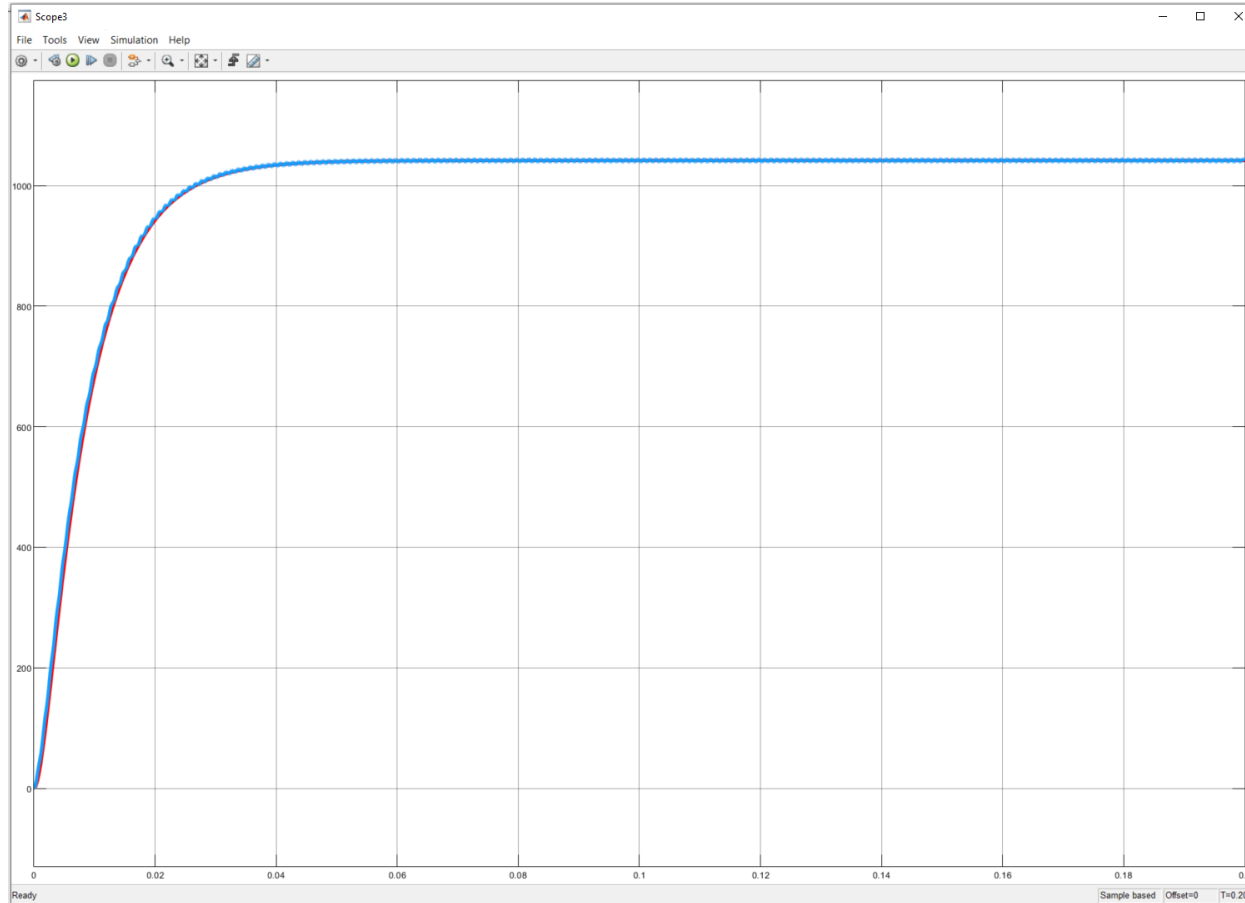
Pulse Width Modulation (PWM)



Period (secs): 0.01

Simulink Stop Simulation: 0.2

Pulse Width Modulation (PWM)

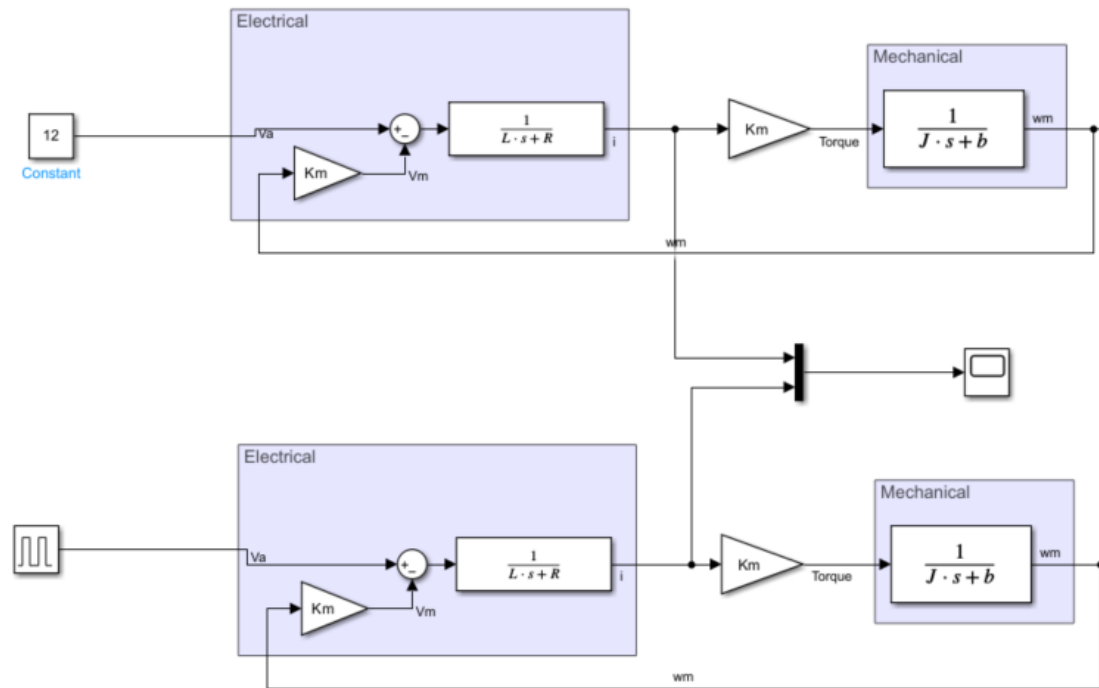


Period (secs): 0.001

Simulink Stop Simulation: 0.2

Pulse Width Modulation (PWM)

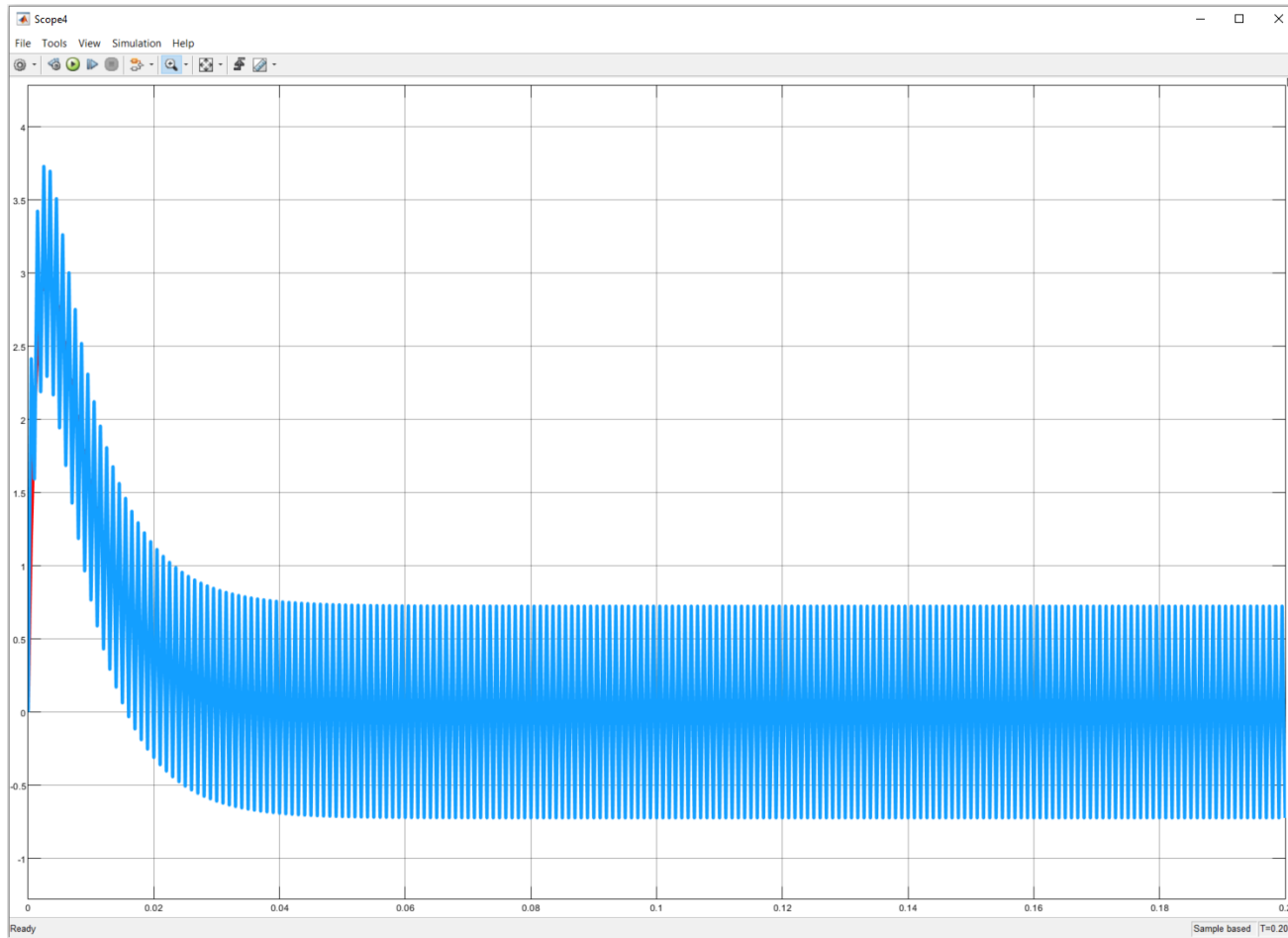
What's about the current?



Fast way for blocks copy

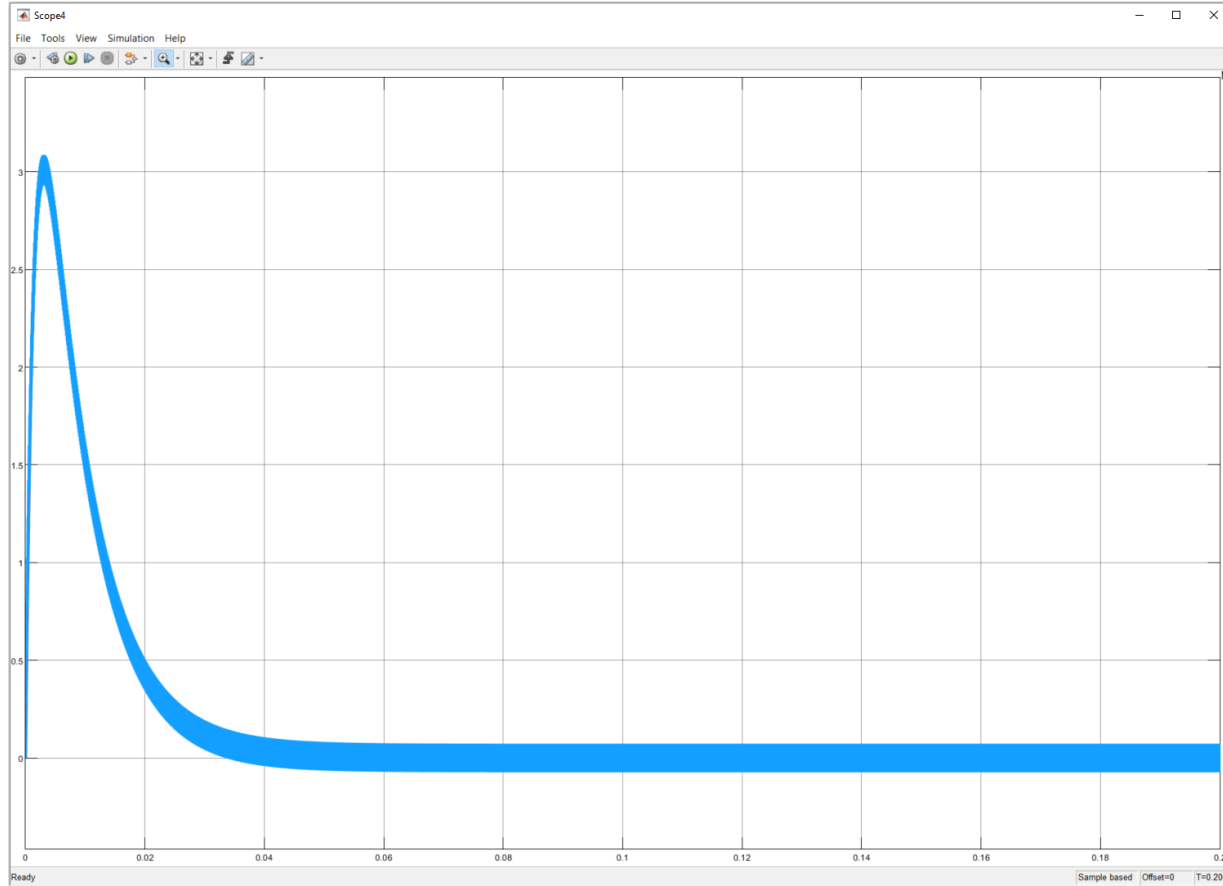
- Select
- CTRL+CLICK
- move

Pulse Width Modulation (PWM)



Inertia on speed

Pulse Width Modulation (PWM)



Period (secs): 0.0001
Simulink Stop Simulation: 0.2

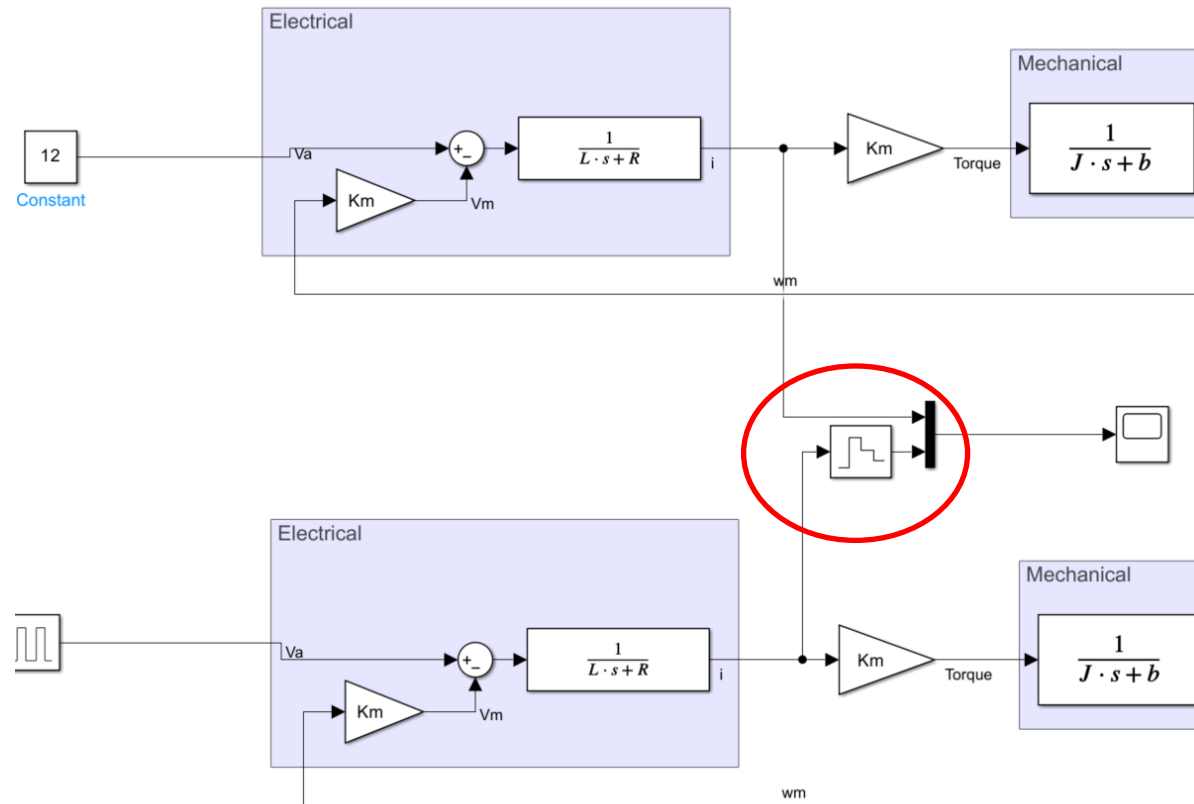
Quite better.
Why don't we use 0.00000.....1?



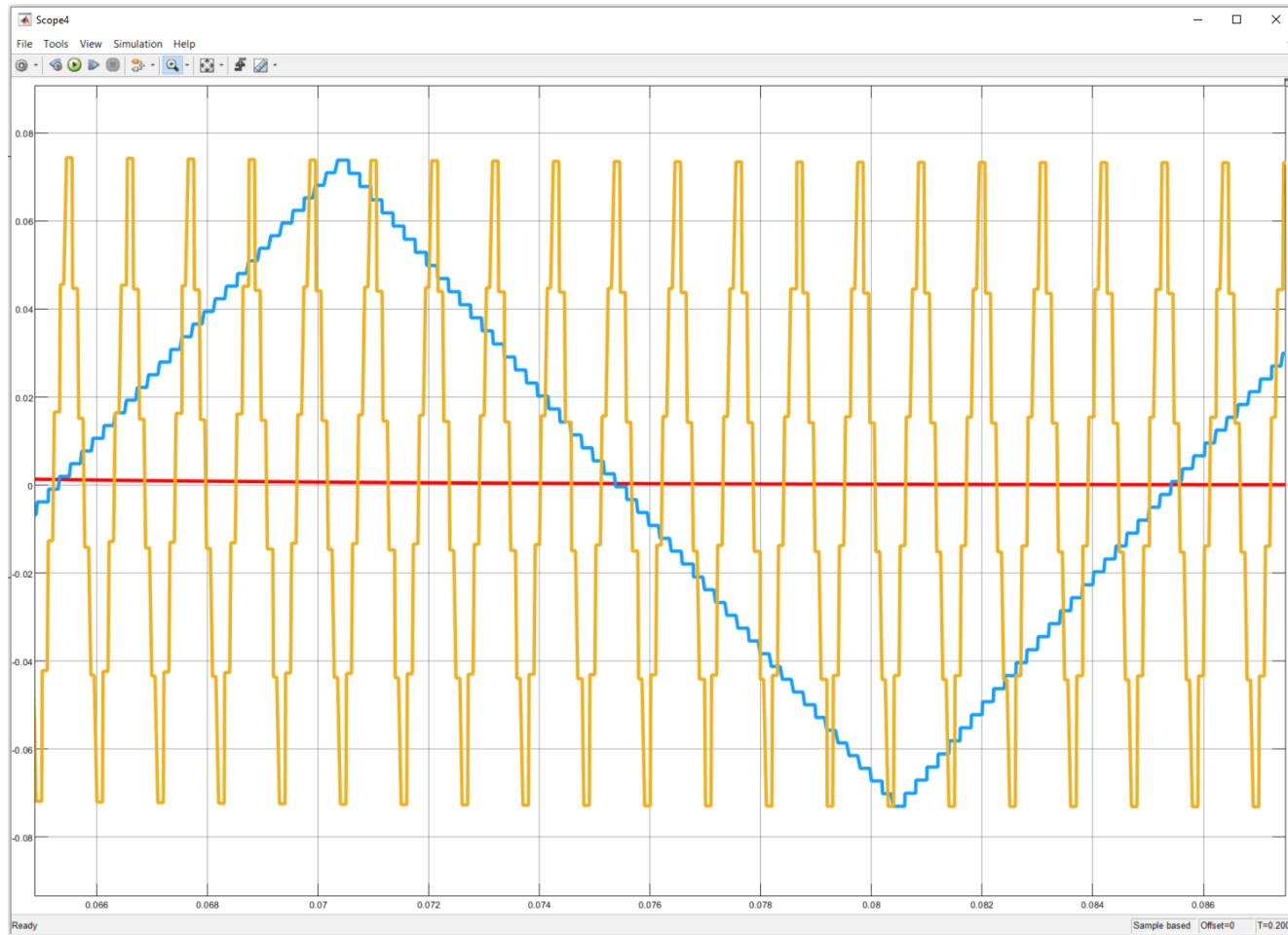
Pulse Width Modulation (PWM)

Another problem...

Try to get samples of the current
at different sample rate



Pulse Width Modulation (PWM)

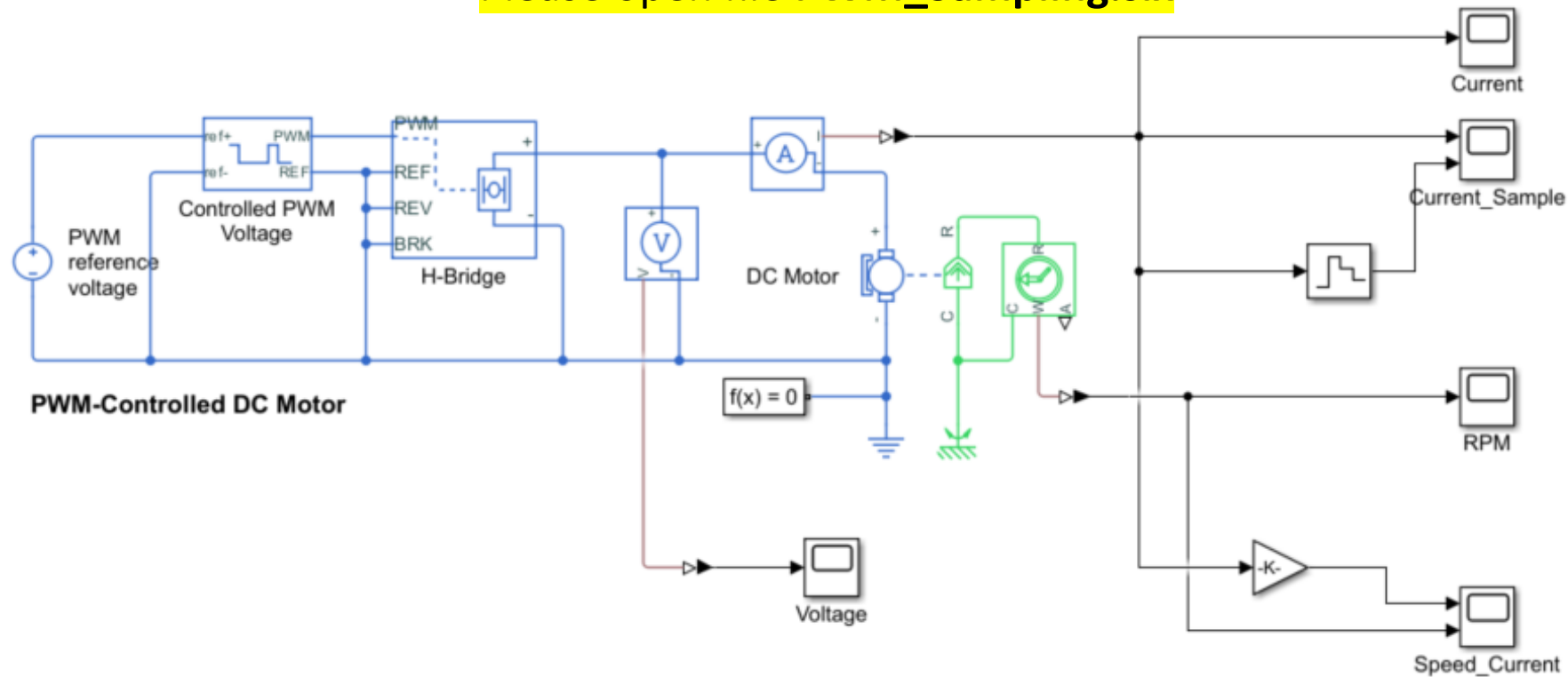


PWM and sampler need a sort of sync

EXERCISE

The main purpose of this exercise is to show you the effect of PWM modulation on the motor current and what do I have to do to have good samples.

Please open file **PWM_Sampling.slx**



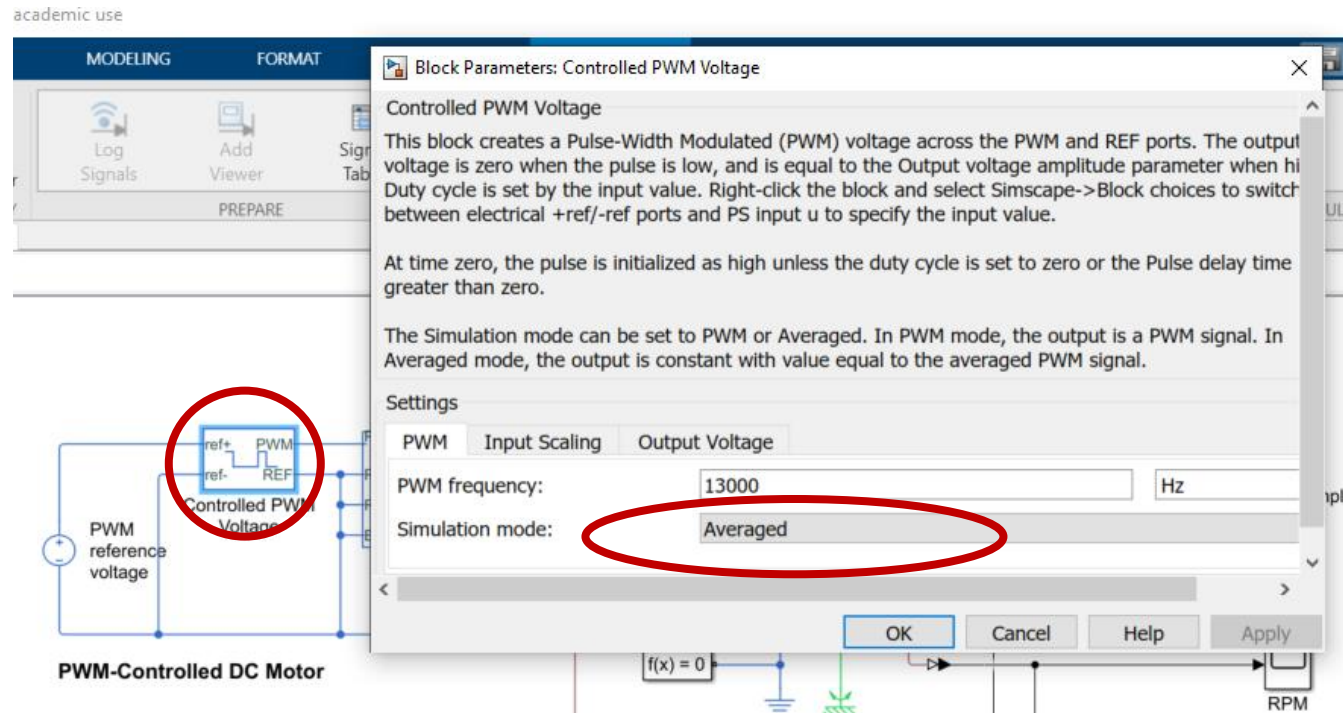
EXERCISE

**Let's start to watching
current with no PWM
(averaged)**

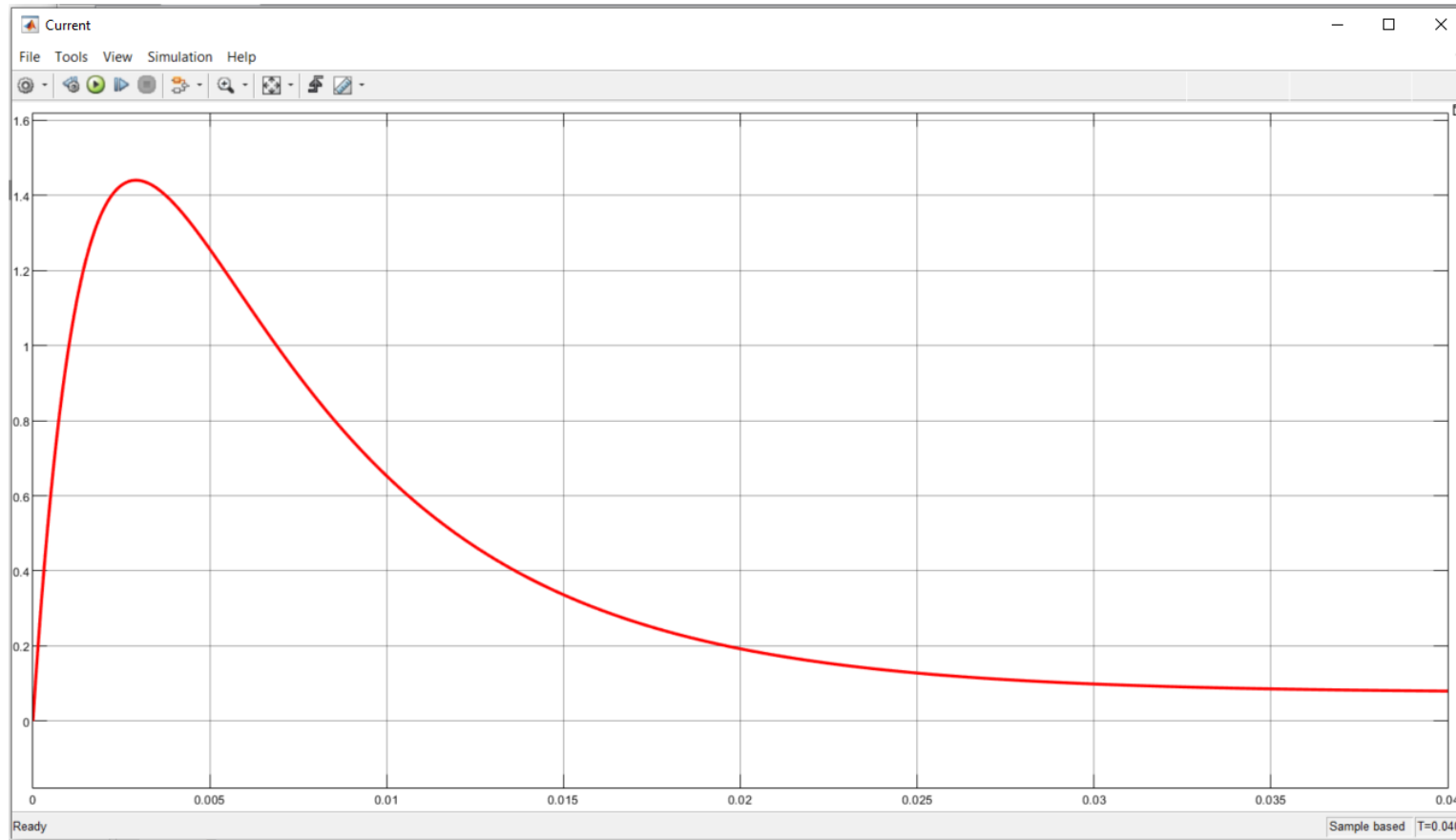
Electrical Time Constant = 1.3 ms

We can choose for example a
Sample Time of 0.1ms = 100us

Tsample=100us

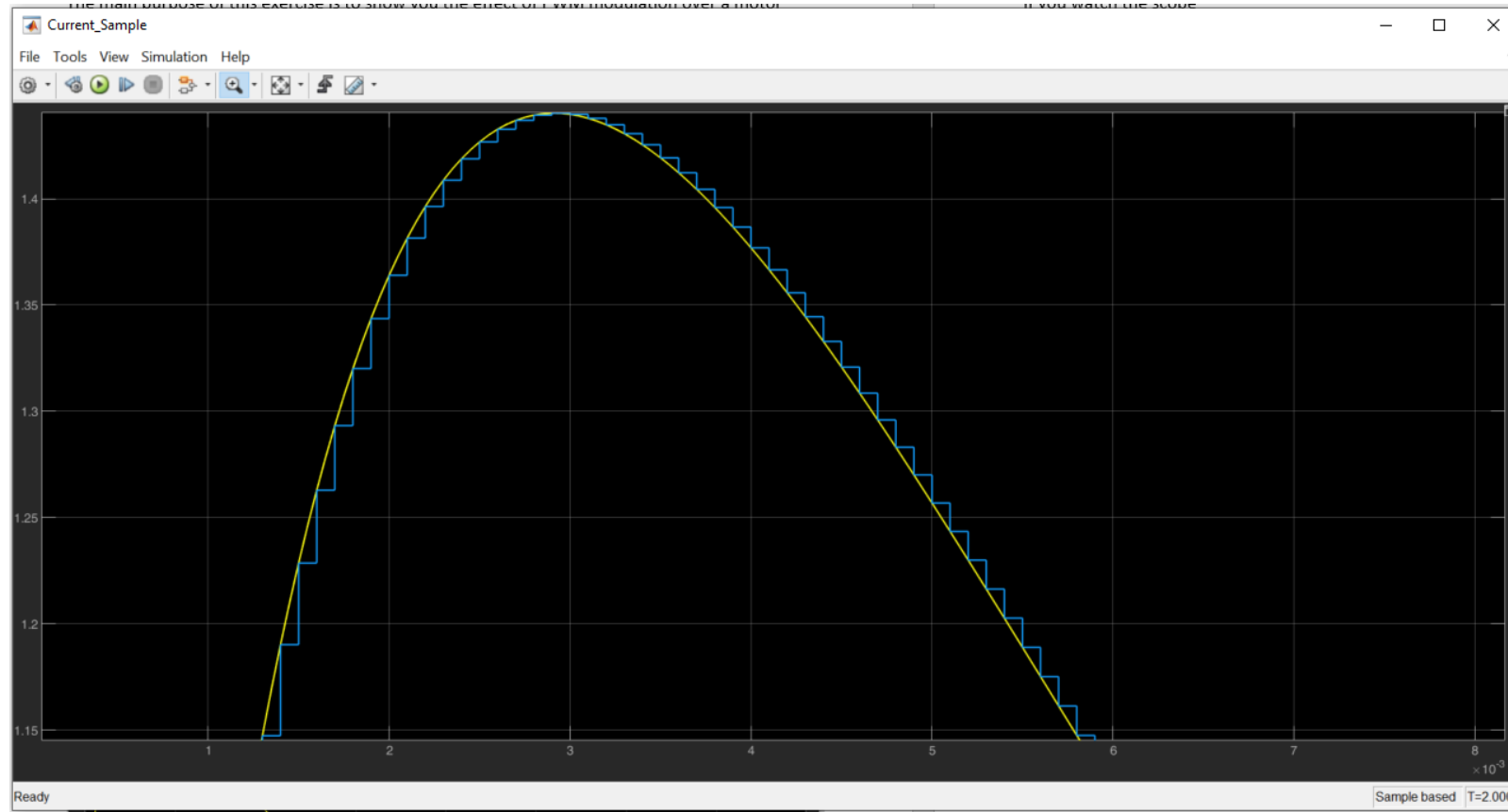


Run the model and watch the scope **Current**



Look the steady state value of the current and keep in mind! (0.08 in my scope)

If you watch **Current_Sample**



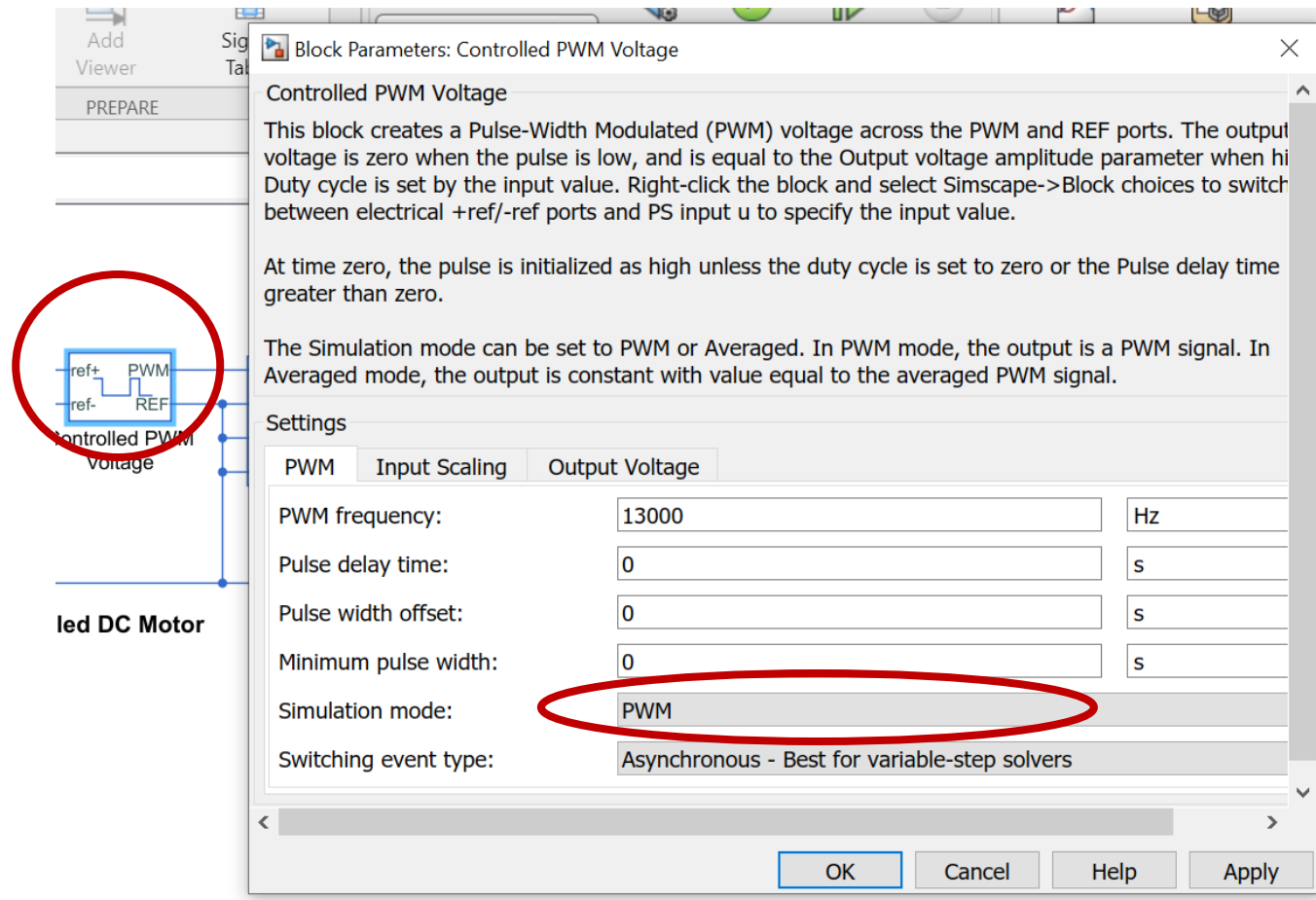
Better Black or White?

You can see:

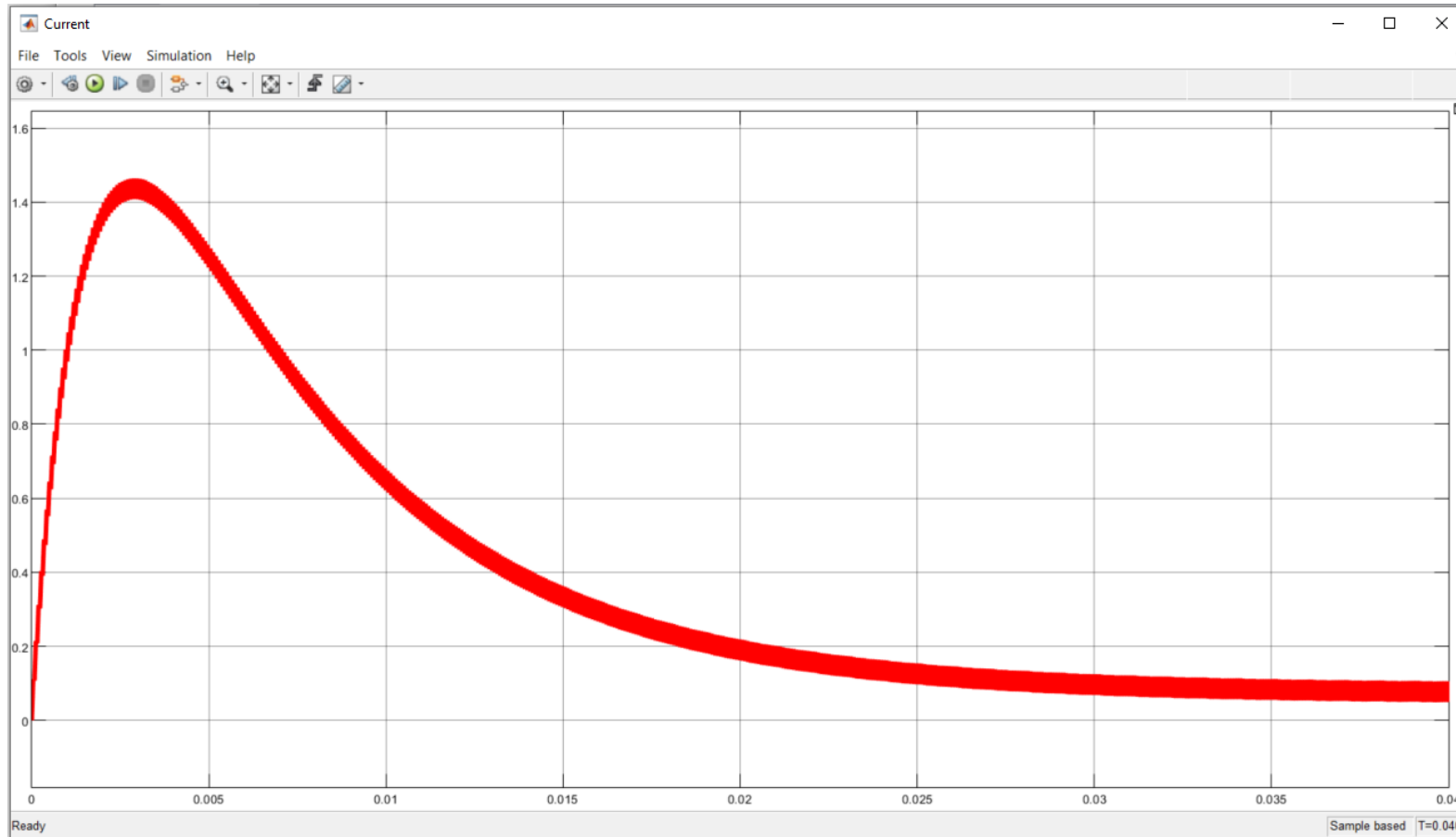
- good approximation in hi-dynamic
- perfect approximation in low-dynamic

Watch voltage scope

Now, let's start the PWM working!

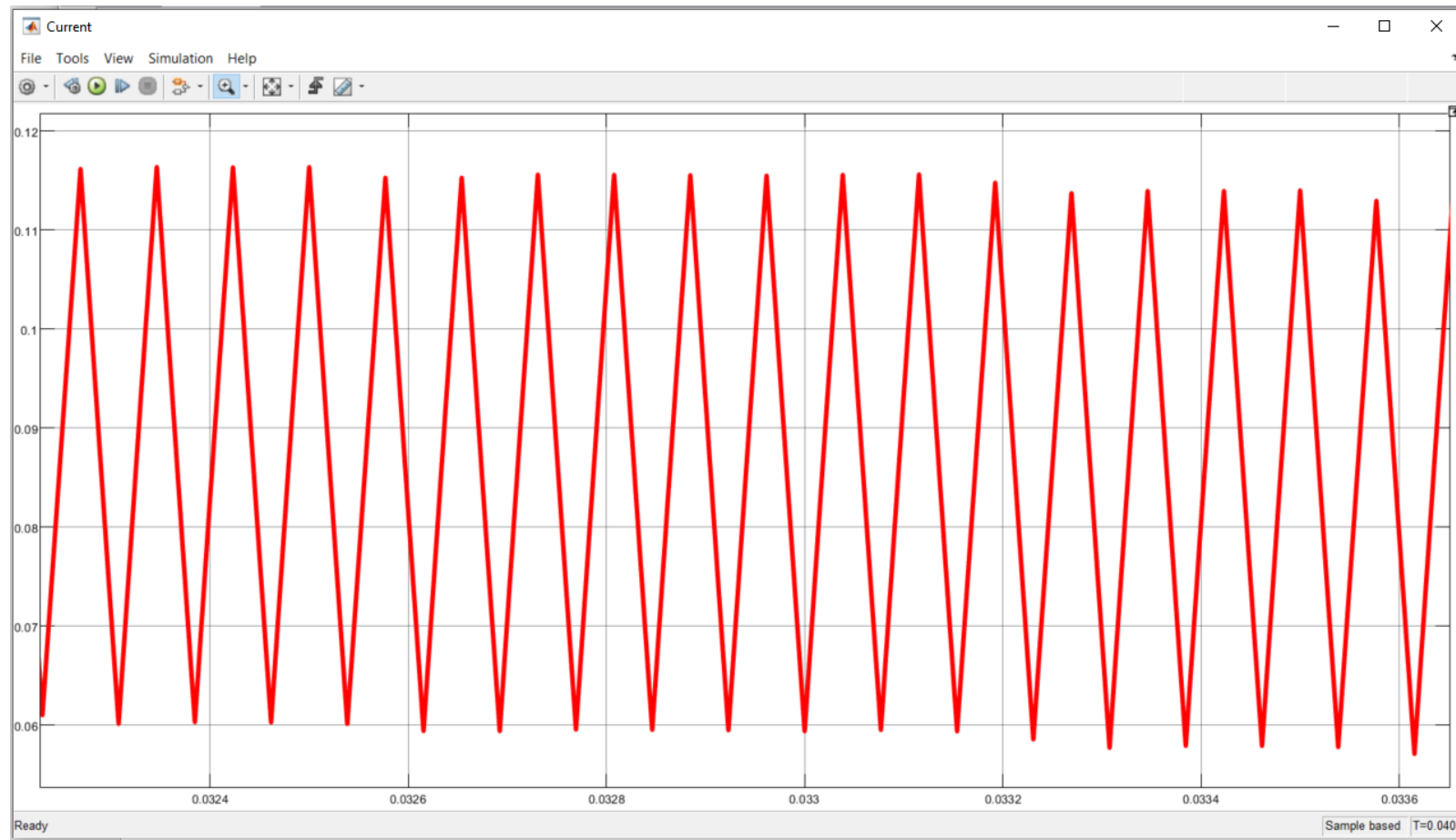


Have a look of the **current** now...

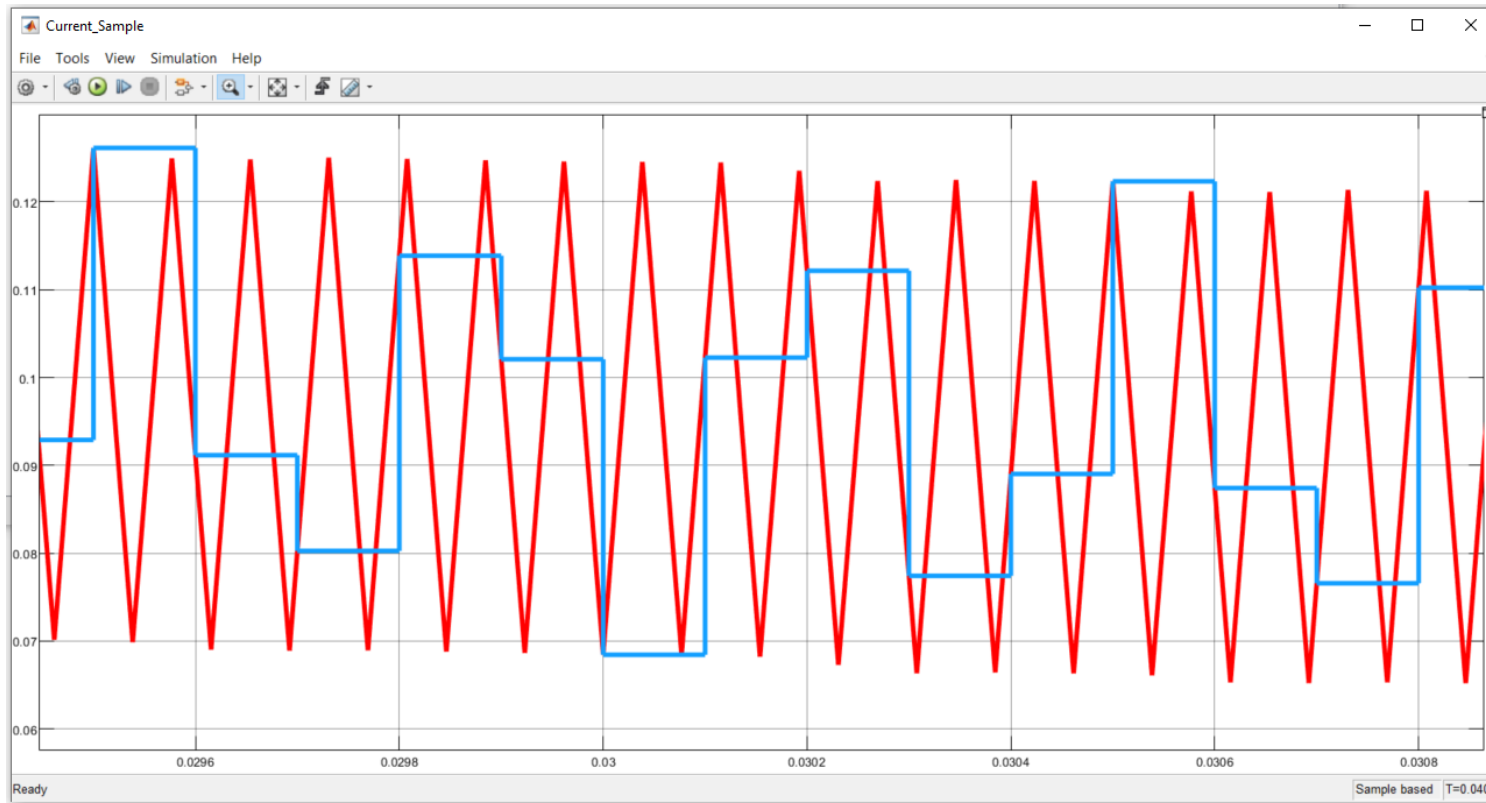


Incredibly “noisy”!

Zoom it!

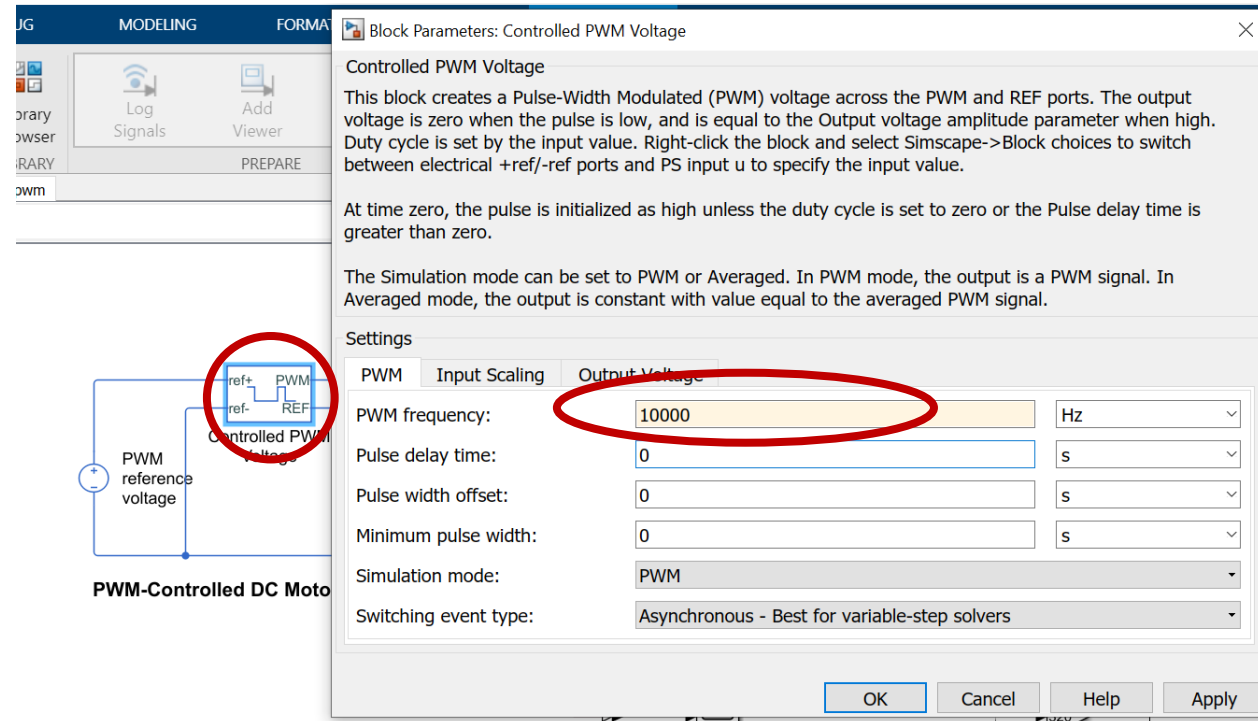


What's about our sampled values?

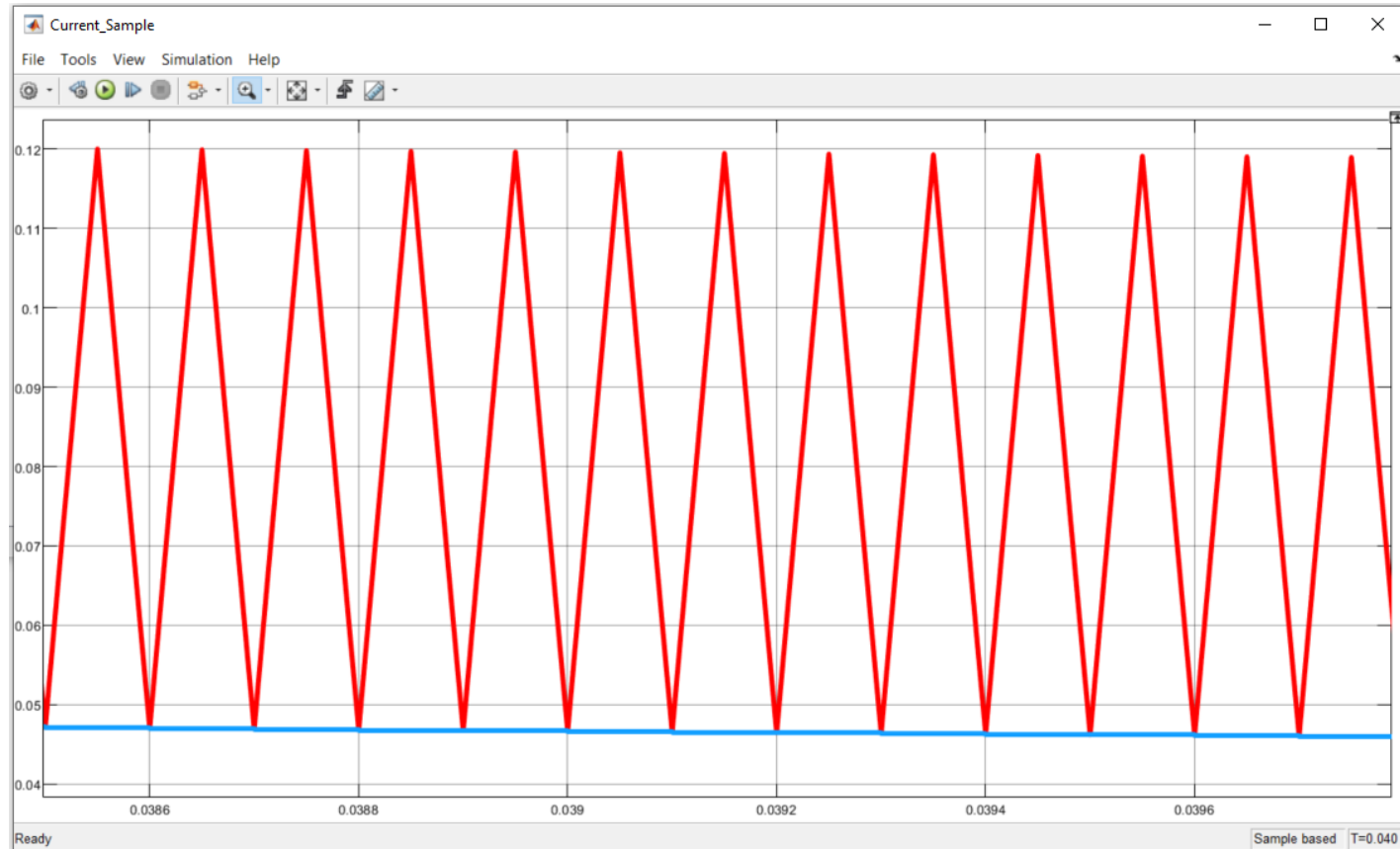


- Do the samples match the signal?
- Do the samples like what we want to feed to our control?

Looking the picture, I think that we can obtain better result if PWM frequency is equal to sampling period (10KHz). Let's try it...

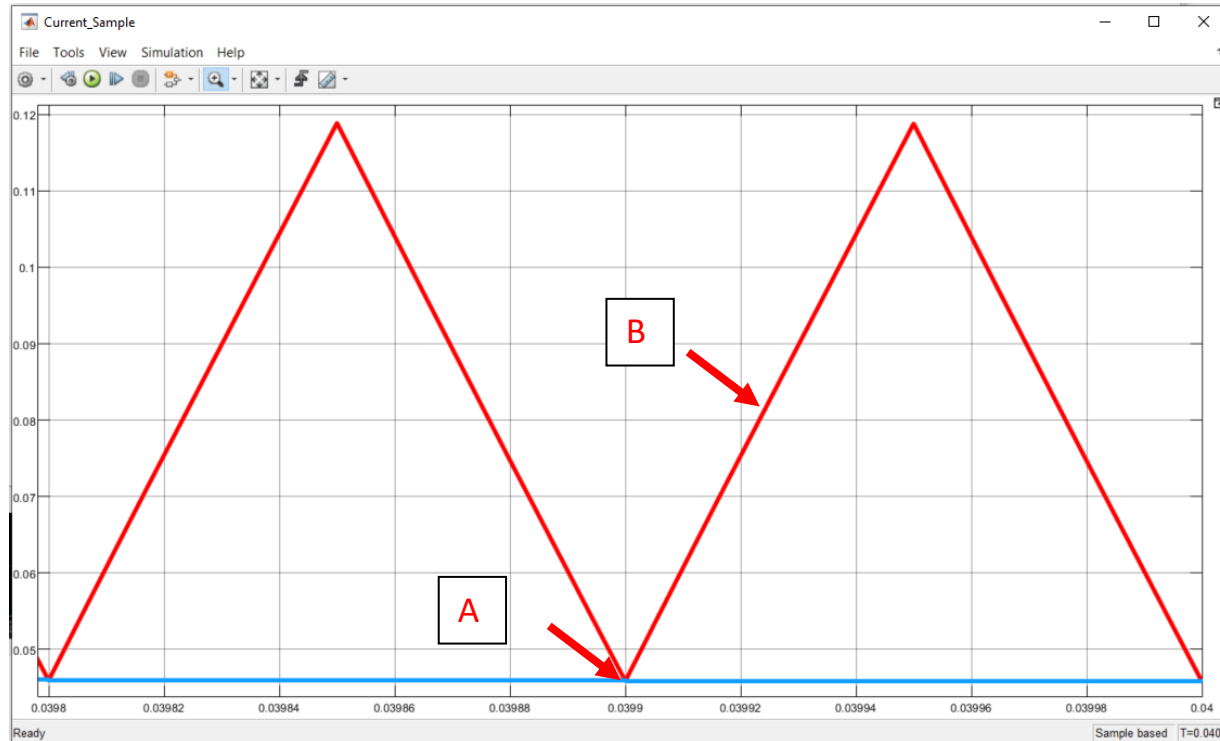


Apply and run simulation.



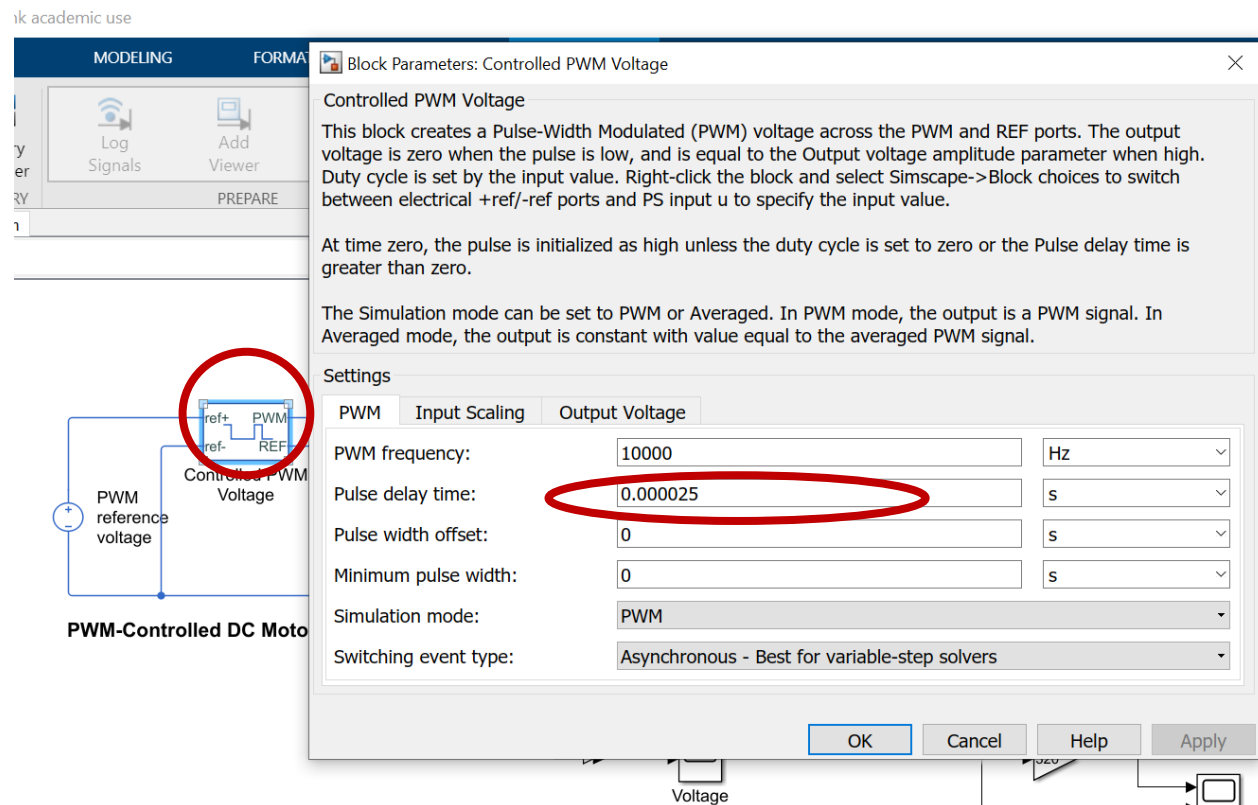
Good! I'm sampling really "noise" signal and I get really good samples!

But is not the value I want; I want the medium value of the triangular wave.
What I have to do is moving my sampling point, from A to B

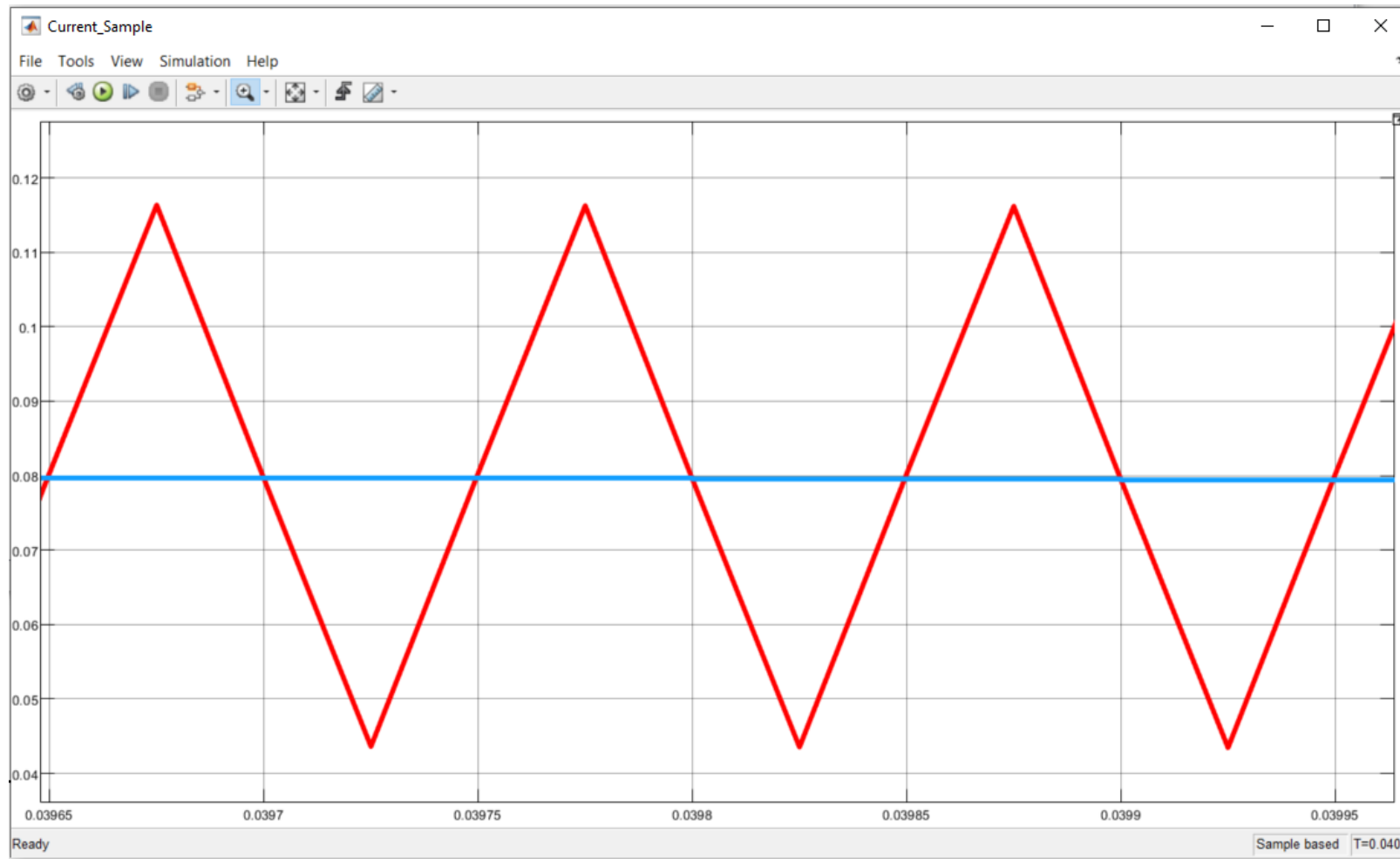


How far is B from A?

It's $\frac{1}{4}$ of the PWM period = $\frac{1}{10\text{KHz}} / 4 = 2.5\text{e-}5$.



And now....



Do you remember the steady state value of the first current? We get it!

Please have a look:

- Voltage with/without PWM
- Try with 1KHz PWM Frequency
- Try with 100KHz PWM Frequency

Why we don't use 100KHz?